Ekikrat: The amalgamation for detecting Hallucination, Ambiguity and Semantic Understanding

Hemangee De*
Roll: 26CS91F02,
CSE
IIT Kharagpur
hemangee4700@gmail.com

Tanushree Dhali*
Roll: 26CS91R08,
CSE
IIT Kharagpur
tanushreedhali000@gmail.com

Debanjan Naskar*
Roll: 26CD91R01,
CCDS
IIT Kharagpur
debanjan989@gmail.com

Abstract

Large language models (LLMs) fail on question answering (QA) in at least three distinct ways: they produce ungrammatical output(semantic understanding), they answer questions that admit several readings as though only one existed (ambiguity), and they state things that are unsupported by, or contradict, the evidence(hallucination). Public resources usually annotate only one of these phenomena. We build a merged corpus of 1,500 question–answer pairs by sampling 500 examples each from AmbigQA, SQuAD 2.0 and HaluBench, and label every example along three axes: grammatical correctness, ambiguity (with its sources), and hallucination. Labels are produced by a reproducible, free pipeline that combines (i) a RoBERTa model fine-tuned on CoLA(Corpus of Linguistic Acceptability) for grammaticality, (ii) the original dataset annotations as ground truth for ambiguity and hallucination wherever they exist, and (iii) an open instruction-tuned LLM (Qwen2.5) as a judge for the remaining rows, with a human-review flag for disagreements. Of the 1,500 examples, 637 contain sentence-length answers that could be scored for grammar (92.5% judged acceptable), 277 of the 500 AmbigQA questions are ambiguous, and 226 of the 500 HaluBench answers are hallucinated. A 30-example pilot of the LLM judge shows that a 3B-parameter judge over-flags ambiguity and frequently abstains on hallucination, which motivates a larger judge and human verification before the corpus is used as a benchmark. We then use the labelled corpus to benchmark four open instruction-tuned models (Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B and Mistral-7B-v0.3, all in 4-bit precision on a free GPU), both as *answerers* (160 questions) and as *judges* of the three properties (up to 277 rows per task). The models answer 68–88% of answerable SQuAD 2.0 questions correctly, yet they hallucinate on 15–35% of unanswerable ones and rarely cover several interpretations of an ambiguous question (mean coverage at most 0.21), and as closed-book hallucination judges they reach an F1 of at most 0.55. Each model has a distinct strength, which makes the corpus a discriminative testbed. We release the pipeline and the labelled file to support further verification. The code and dataset can be found in : `https://github.com/Tanushree100/Ekikrat`.

1 Introduction

Question answering is one of the most common ways in which users interact with LLMs, and it is also where their weaknesses become visible. An answer can be wrong because it is *malformed* (not grammatical), because the question itself is *under-specified* (several valid interpretations exist), or because the model *hallucinates*, producing content that is not supported by the source material or by the world (Ji et al. 2023; Huang et al. 2025). These three failure modes are usually studied in isolation: ambiguity in AmbigQA (Min et al. 2020), unanswerability in SQuAD 2.0 (Rajpurkar et al. 2018), and hallucination in benchmarks such as HaluEval, RAGTruth and HaluBench (Li et al. 2023; Niu et al. 2024; Ravi et al. 2024). As a result, there is no single resource in which the same example carries labels for all three, which makes it hard to ask whether, for instance, ambiguous questions are more likely to elicit hallucinated answers.

This report describes the construction of such a resource. We make four contributions:

- **A merged corpus.** We draw a seeded random sample of 500 examples from each of AmbigQA, SQuAD 2.0 and HaluBench, normalise them to a common schema and shuffle them, giving 1,500 examples.
- **A three-axis labelling procedure.** Each example receives a grammatical-correctness label, an ambiguity label with ambiguity sources, and a hallucination label, together with the basis for the label (dataset annotation or model judgement) and a flag for human review.
- **An honest account of label quality.** We report label statistics and a small pilot of the LLM judge, and we state clearly which labels are dataset ground truth, which are automatic, and which have not yet been verified by human annotators.
- **A four-model benchmark on the corpus.** We evaluate Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B and Mistral-7B-v0.3 in two roles: answering the questions (hallucination and ambiguity handling) and judging the three labels, and we show that the corpus separates the models' strengths and weaknesses.

2 Literature Review

Hallucination. Surveys define hallucination in natural language generation as content that is nonsensical or unfaithful to the provided source, and distinguish *intrinsic* hallucinations that contradict the source from *extrinsic* ones that cannot be verified from it (Ji et al. 2023; Maynez et al. 2020). More recent surveys extend this taxonomy to LLMs and separate faithfulness hallucinations from factuality hallucinations (Huang et al. 2025). TruthfulQA measures whether models reproduce common human misconceptions (Lin et al. 2022); HaluEval supplies large collections of generated and human-annotated hallucinated answers (Li et al. 2023); and RAGTruth provides word-level hallucination annotations for retrieval-augmented generation (Niu et al. 2024). HaluBench aggregates samples from several of these and from QA datasets (DROP, CovidQA, PubMedQA, FinanceBench), pairing each context–question–answer triple with a PASS/FAIL faithfulness label, and was introduced together with the Lynx evaluator (Ravi et al. 2024; Dua et al. 2019; Möller et al. 2020; Jin et al. 2019; Islam et al. 2023).

Hallucination detection. Detection methods range from reference-based checks against a context to reference-free consistency checks. SelfCheckGPT samples several responses and treats disagreement among them as evidence of hallucination (Manakul et al. 2023), and semantic entropy estimates uncertainty over meanings rather than token sequences, thereby accounting for linguistic invariances (Kuhn et al. 2023). Using a strong LLM as an evaluator is now common practice, and its agreement with human judgement has been studied systematically (Zheng et al. 2023).

Ambiguity and unanswerability. Many open-domain questions are ambiguous. AmbigQA studies this on questions from Natural Questions (Kwiatkowski et al. 2019): annotators mark a question as ambiguous when it has multiple distinct answers and rewrite it into disambiguated question–answer pairs, each representing one reading (Min et al. 2020). SQuAD 2.0 adds questions that cannot be answered from the given paragraph, forcing models to abstain (Rajpurkar et al. 2018), building on the original SQuAD (Rajpurkar et al. 2016). The two phenomena are related but not identical: in AmbigQA the problem lies in the question, while in SQuAD 2.0 it lies in the lack of supporting evidence.

Grammatical acceptability. CoLA introduced the task of judging whether a sentence is acceptable to a native speaker (Warstadt et al. 2019). Fine-tuning pretrained encoders such as RoBERTa (Liu et al. 2019) on CoLA yields strong acceptability classifiers, and a RoBERTa-base model fine-tuned on CoLA is distributed with the TextAttack library (Morris et al. 2020). We use it as an inexpensive grammar scorer.

Open LLMs as annotators. Instruction-tuned open models such as the Qwen2.5 family (Qwen Team 2024) can be run without API cost on a single free GPU, which makes them attractive as first-pass annotators, although smaller models are known to be noisier judges than frontier models (Zheng et al. 2023). Our pipeline therefore treats dataset annotations as primary evidence and the LLM as a cross-check. In the benchmark of Section 3.3 we additionally evaluate Llama-3.1-8B (Dubey et al. 2024), Gemma-2-9B (Gemma Team 2024) and Mistral-7B (Jiang et al. 2023) next to Qwen2.5-7B, all loaded with 4-bit NormalFloat quantisation (Dettmers et al. 2023) so that every model fits on a free 16 GB GPU.

3 Methodology

3.1 Data Collection

3.1.1 Data Preparation

We use three public sources, loaded with the Hugging Face `datasets` and `transformers` libraries (Lhoest et al. 2021; Wolf et al. 2020).

AmbigQA (`ambig_qa`, "light" configuration, validation split) is open-domain, so no passage is available. Each question is annotated either as `singleAnswer` or `multipleQAs`. For single-answer questions we store the gold answer(s); for multiple-QA questions we store all disambiguated answers joined with "`|`" and keep the disambiguated question–answer pairs as a JSON string.

SQuAD 2.0 (validation split) provides a paragraph, a question and zero or more gold answer spans. An empty answer list marks an unanswerable question; we record this in a reference column and leave the answer empty.

HaluBench (test split) provides a context, a question, an answer and a PASS/FAIL label, plus a field naming the source dataset (e.g., HaluEval, DROP).

From each source we draw 500 rows uniformly at random with seed 42.

3.1.2 Merging of the Datasets

All three samples are mapped to one schema: `id`, `source`, `context`, `question`, `answer`, and the reference columns `ref_original_label`, `ref_ambigqa_type`, `ref_ambigqa_disambiguated` and `ref_squad_unanswerable`, which preserve the original annotations. The samples are concatenated and shuffled (seed 42) to remove ordering by source. Empty annotation columns are added for the three labels, a free-text `ambiguity_sources` field, an annotator identifier and a notes field, so that the file can also

be annotated manually in a spreadsheet with Yes/No drop-downs. Table 1 and Figure 1 give the composition of the merged corpus, including the HaluBench source mix and the share of sentence-level, short-span and empty answers. Answers are short: the median is 3 words (mean 7.9), and 277 answers (all SQuAD 2.0 unanswerables) are empty.

Table 1: Composition of the merged corpus (1,500 examples). HaluBench is split by its source dataset.

Source	Split used	Examples
AmbigQA (light)	validation	500
SQuAD 2.0	validation	500
HaluBench	test	500
HaluEval		343
FinanceBench		37
PubMedQA		32
RAGTruth		31
DROP		30
CovidQA		27
Total		1,500

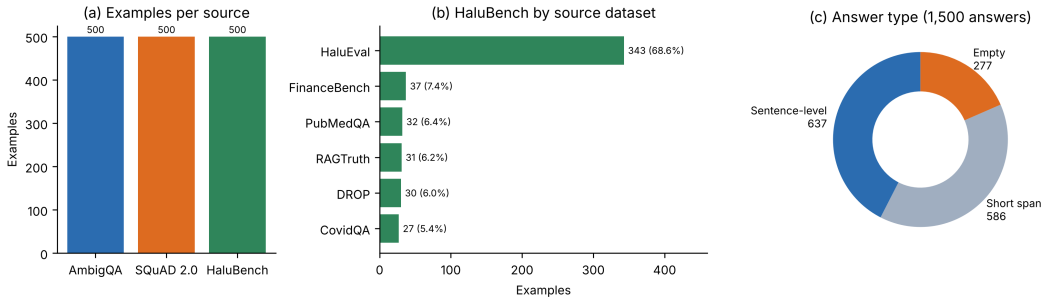


Figure 1: Composition of the merged corpus (Ekikrat). (a) Each source contributes 500 examples. (b) HaluBench examples by source dataset; HaluEval-derived rows make up 68.6% of HaluBench. (c) Answer types over all 1,500 rows: 637 sentence-level answers that can be scored for grammar, 586 short spans (at most three words, no terminal punctuation) and 277 empty answers (all SQuAD 2.0 unanswerable questions).

3.1.3 Labelling into Three Categories: Grammar, Ambiguity, Hallucination

Every example is described by three labels, each in $\{\text{Yes}, \text{No}\}$ with N/A where the label is undefined:

- **Grammatical correctness** (`auto_grammar_label`): whether the *answer* is a grammatically acceptable text.
- **Ambiguity** (`auto_ambiguous`, `auto_ambiguity_sources`): whether the *question* admits more than one reasonable interpretation, or the answer is vague or lists competing candidates, and if so why.
- **Hallucination** (`auto_halluc_label`): whether the *answer* contradicts or is unsupported by the context or, when there is no context, is factually wrong.

Each label is accompanied by its basis (`dataset_label`, `llm` or `empty_answer`) and a review flag. Table 2 summarises where each label comes from.

Table 2: Provenance of the labels for each source.

Source	Grammar	Ambiguity	Hallucination
AmbigQA	CoLA model	dataset (`multipleQAs`)	gold answer ⇒ No
SQuAD 2.0	CoLA model	LLM judge	gold answer ⇒ No
HaluBench	CoLA model	LLM judge	dataset (FAIL ⇒ Yes)

3.2 Procedure of Labelling

3.2.1 Data Labelling Algorithm

Grammar. Answers are first cleaned (the "|" separator of multi-answer AmbigQA rows becomes a comma). Empty answers are N/A. Answers of at most three words that do not end in terminal punctuation are treated as *short spans* (e.g., "Denmark", "1998"); these are fragments rather than sentences, so a sentence-level acceptability model cannot judge them fairly and they are labelled N/A. All other answers are split into sentences, each sentence is scored by `textattack/roberta-base-CoLA` (Morris et al. 2020), and the answer score is the mean of its sentence scores. An answer is labelled Yes if the mean is at least 0.5, and flagged for human review if the mean lies in $[0.3, 0.7]$. As published descriptions of the label order of this checkpoint disagree, the code verifies at run time which output index denotes "acceptable" (Section 3.2.2).

Ambiguity and hallucination. Dataset annotations take priority. For AmbigQA, `multipleQAs` means ambiguous, and the ambiguity sources are the disambiguated question–answer readings. For HaluBench, FAIL means hallucinated and PASS means not. Gold SQuAD 2.0 and AmbigQA answers are written by annotators and are therefore labelled not hallucinated. Rows without a dataset label (ambiguity for SQuAD 2.0 and HaluBench) are judged by `Qwen2.5-3B-Instruct` (Qwen Team 2024) prompted with a system message containing two worked examples and instructed to return a single JSON object with fields `is_ambiguous`, `ambiguity_sources`, `is_hallucinated` (Yes/No/Unsure) and `hallucination_reason`. Decoding is greedy, contexts are truncated to 3,000 characters, results are cached to disk so that a disconnected session can resume, and malformed outputs are recovered with a regular-expression fallback. The LLM verdict on hallucination is also computed for rows that have dataset labels, but it only triggers a review flag when it disagrees. Algorithm 1 gives the procedure.

3.2.2 Model Testing Algorithm

Because both automatic judges are imperfect, each is tested before and after use (Algorithm 2).

Grammar model check. Three grammatical and three deliberately scrambled sentences are scored; the label index whose mean probability is higher on the grammatical set is taken to mean "acceptable", and the run aborts if the gap between the two means is below 0.2.

Judge check against dataset labels. Wherever a dataset label exists, the LLM verdict is compared with it by cross-tabulation: HaluBench PASS/FAIL against the LLM hallucination verdict; SQuAD 2.0 gold answers (which should be judged not hallucinated); and AmbigQA `singleAnswer`/`multipleQAs` against the LLM ambiguity verdict. In addition, the notebook has a smoke-test mode that judges only 30 random rows before a full run, and it counts unparseable outputs. Rows on which any check disagrees are flagged for human review.

3.3 Benchmarking Open LLMs on the Labelled Corpus

Beyond serving as a labelled resource, the corpus can be used to test models. We run two complementary tests on four instruction-tuned models: `Llama-3.1-8B-Instruct`, `Qwen2.5-7B-Instruct`, `Gemma-2-9B-it` and `Mistral-7B-Instruct-v0.3`. Each model is loaded once in 4-bit NF4 precision with fp16 compute on a free Colab T4 GPU, and

Algorithm 1 Three-axis data labelling

Require: Merged rows (id, src, c, q, a) with reference fields

for each row **do**
  if a is empty **then**
    grammar, ambiguity, hallucination ← N/A; **continue**
  end if
  $t \leftarrow$ clean(a)
  if $|t| \leq 3$ words and no terminal punctuation **then**
    grammar ← N/A
  else
    $s \leftarrow$ mean CoLA acceptability over sentences of t
    grammar ← Yes if $s \geq 0.5$ else No; review ← $[0.3 \leq s \leq 0.7]$
  end if
  $j \leftarrow$ LLMJudge(q, c, a) ▷ cached; JSON output
  if src = AmbigQA **then**
    ambiguity ← ($type$ = `multipleQAs`); hallucination ← No
  else if src = SQuAD 2.0 **then**
    ambiguity ← j.ambiguous; hallucination ← No
  else
    ambiguity ← j.ambiguous; hallucination ← (label = FAIL)
  end if
  review ← review ∨ (j contradicts a dataset label) ∨ (j.halluc = Unsure)
end for

Algorithm 2 Model testing

Score sets G (grammatical) and B (scrambled) with the CoLA model
acceptable ← label index with larger mean on G than on B; **assert** gap > 0.2
Run LLMJudge on a random pilot subset (30 rows); count unparseable outputs
for each judged row with a dataset label **do**
  update cross-tabulation (dataset label × LLM verdict)
  if verdicts disagree **then**
    flag row for human review
  end if
end for
return agreement tables and review list

decoding is greedy with batched generation. All sampling uses seed 0 and 40 rows per group; the same sample is used for every model. The Llama-3.1 checkpoint is gated, so it was run in a separate, otherwise identical session of the same notebook.

Test A: the model answers. Four groups of 40 questions are drawn: AmbigQA `singleAnswer`, AmbigQA `multipleQAs`, SQuAD 2.0 answerable and SQuAD 2.0 unanswerable (160 questions in total). Questions with a passage (SQuAD 2.0) are answered "using only the context", with the exact reply `UNANSWERABLE` allowed (48 new tokens). Open-domain AmbigQA questions are answered in a few words; the model may reply `AMBIGUOUS:` followed by each interpretation and its answer, or `UNKNOWN` (120 new tokens). Predictions and gold answers are normalised (lower-cased, punctuation and articles removed), and a prediction is correct if it contains a gold answer or has token-level $F_1 \geq 0.5$ with one. We then report:

- the *hallucination rate* on unanswerable questions (the model answered instead of abstaining) and the *wrong-answer rate* on answerable ones (the model answered, but incorrectly);
- for `multipleQAs` questions, the share of answers that *flag ambiguity* (start with `AMBIGUOUS`), the *mean interpretation coverage* (the fraction of the disambiguated

question–answer pairs matched by the prediction), and the share that gives at least one valid answer;

- for `singleAnswer` questions, the *false ambiguity flag rate*, so that a high flag rate on multi-answer questions can be read against it.

Test B: the model judges. The same models label rows of the corpus with a one-word Yes/No prompt (4 new tokens), and their verdicts are scored against the corpus labels (`auto_halluc_label`, `auto_ambiguous`, `auto_grammar_label`). Hallucination is judged from the context when there is one and otherwise from the model's own knowledge; ambiguity is judged from the question alone; "understandability" asks whether the *answer text* is grammatically correct and easy to understand. Up to 40 rows are drawn per source, among rows that carry the label in question (Table 3). The positive class is Yes for hallucination and ambiguity and No (a grammar or clarity problem) for understandability; we report accuracy and precision, recall and F_1 of the positive class, as well as the fraction of unparseable outputs.

Table 3: Test sets used for the benchmark.

Test	Rows	Composition
A: answering	160	4 groups $\times$ 40 (AmbigQA single/multi, SQuAD 2.0 answerable/unanswerable)
B: hallucination judge	277	$\leq$40 per source: AmbigQA, SQuAD 2.0 and six HaluBench sources
B: ambiguity judge	68	40 AmbigQA rows with dataset labels + 28 rows with LLM-pilot labels
B: understandability judge	217	$\leq$40 per source, among rows with a grammar label

Passage re-attachment. The notebook tries to re-attach the original HaluBench passages from the public dataset before testing. In our runs this lookup failed (a missing-field error), so HaluBench rows were judged closed-book. We therefore read the HaluBench part of the hallucination judge test as a closed-book fact-checking test, not as a test of faithfulness to a passage.

4 Results

Broad label coverage. The corpus of 1,500 rows is richly annotated. A total of 1,223 rows receive a hallucination label directly from published dataset annotations, and the 277 rows with empty SQuAD 2.0 answers are principled N/A cases. All 500 AmbigQA rows carry dataset-based ambiguity labels, and an initial LLM-judge run adds ambiguity labels for 28 further rows. Figure 2 summarises this coverage for the three axes. The ambiguity axis currently has 500 dataset-based labels and 28 LLM-pilot labels; 695 non-empty rows are waiting for the full judge run, and 277 rows are N/A because their answer is empty. The manual columns (`is_grammatically_correct`, `is_ambiguous`, `is_hallucinated`) are ready for human annotation, giving the corpus a clear path to a fully validated release.

Grammar. Table 4 and Figure 3 show that the grammar label cleanly separates sentence-level answers (637) from empty answers and short spans (863). Of the sentence-level answers, 589 (92.5%) are judged acceptable. The mean answer score is 0.825 (median 0.908), and 112 answers fall in the review band, which is a ready-made queue for human checking. Inspecting the lowest-scoring answers is informative in itself: it surfaces span-like fragments and concatenated AmbigQA answers, so the label captures *sentence acceptability* and offers a useful lens on answer form in QA data.

Ambiguity. AmbigQA contributes a substantial ambiguous subset: 277 of 500 questions (55.4%) are `multipleQAs` and 223 are `singleAnswer`, giving a balanced base for studying ambiguity (Figure 4a). In the 30-row LLM pilot, the judge returned parseable verdicts for 28 rows. It flagged all 10 sampled SQuAD 2.0 questions as ambiguous (Figure 4b), which is implausible for questions

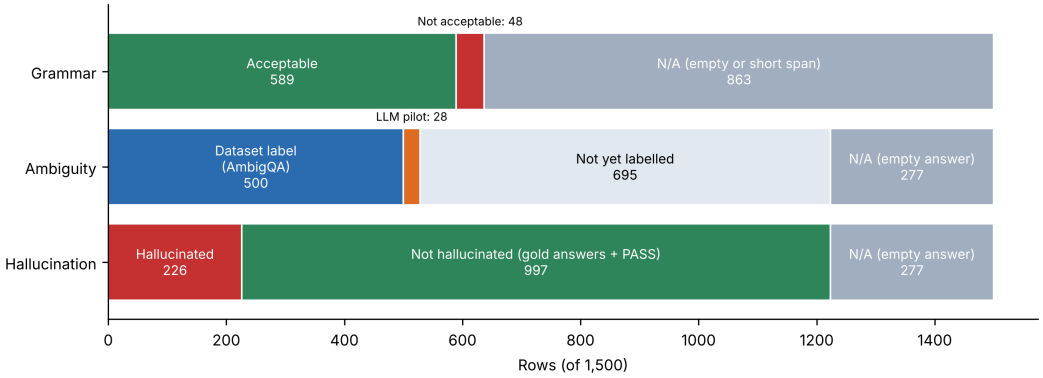


Figure 2: Label coverage of the corpus on the three axes (1,500 rows each). Grammar: 637 rows are scored and 863 are N/A (277 empty and 586 short spans). Ambiguity: 500 AmbigQA rows carry dataset labels, 28 rows carry LLM-pilot labels, 695 non-empty rows are not yet labelled and 277 are N/A. Hallucination: 226 rows are hallucinated (HaluBench FAIL) and 997 are not (HaluBench PASS and gold SQuAD 2.0/AmbigQA answers); 277 are N/A.

Table 4: Grammar labels by source (sentence-level answers only; N/A = empty answer or short span).

Source	Yes	No	N/A	% Yes (scored)
AmbigQA	280	16	204	94.6
SQuAD 2.0	53	10	437	84.1
HaluBench	256	22	222	92.1
Total	589	48	863	92.5

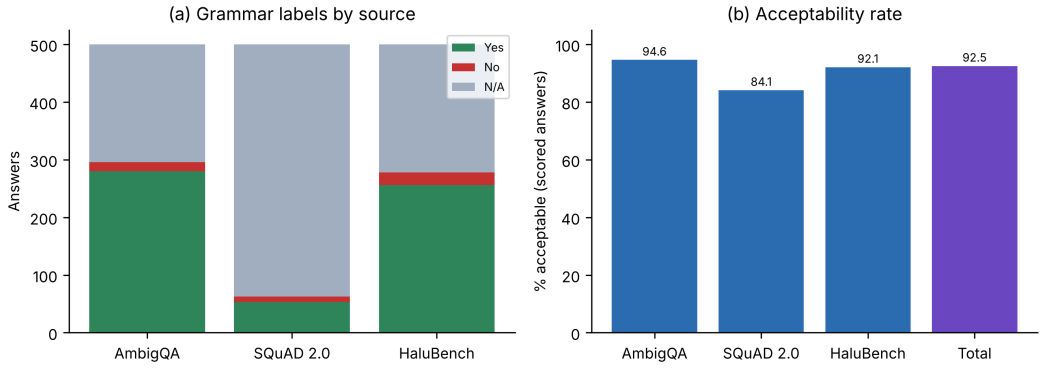


Figure 3: Grammar labels from the CoLA model. (a) Number of answers labelled acceptable (Yes), not acceptable (No) or N/A for each source. (b) Percentage of scored (sentence-level) answers judged acceptable: 94.6% (AmbigQA), 84.1% (SQuAD 2.0), 92.1% (HaluBench) and 92.5% overall.

written against a specific paragraph. This sanity check shows that a small judge is over-sensitive to ambiguity, and it led us to report LLM-based ambiguity rates only where they are trustworthy (not for SQuAD 2.0 or HaluBench).

Hallucination. HaluBench contributes a well-balanced mix of hallucinated and faithful answers: 226 (45.2%) are FAIL (hallucinated) and 274 PASS. Table 5 and Figure 5 break this down across

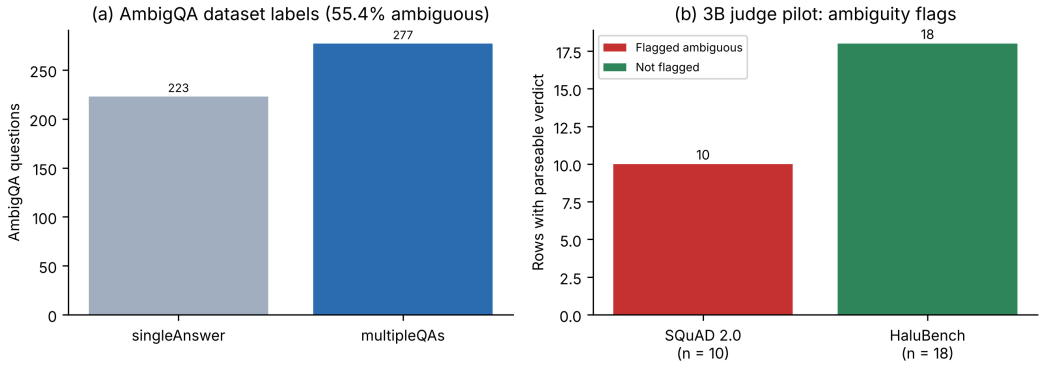


Figure 4: Ambiguity labels. (a) AmbigQA dataset labels: 277 of 500 questions (55.4%) are `multipleQAs`. (b) Verdicts of the 3B LLM judge in the 30-row pilot (28 parseable): all 10 SQuAD 2.0 questions were flagged ambiguous, which is implausible and indicates over-flagging; the 18 parseable HaluBench rows were not flagged.

six source datasets, from 19.4% (RAGTruth) to 56.7% (DROP), so the corpus covers a wide range of hallucination rates.

Table 5: Hallucination labels in HaluBench by source dataset (dataset FAIL = hallucinated).

Source	Hallucinated	Not hallucinated	% hallucinated
HaluEval	157	186	45.8
FinanceBench	19	18	51.4
PubMedQA	17	15	53.1
DROP	17	13	56.7
CovidQA	10	17	37.0
RAGTruth	6	25	19.4
HaluBench total	226	274	45.2

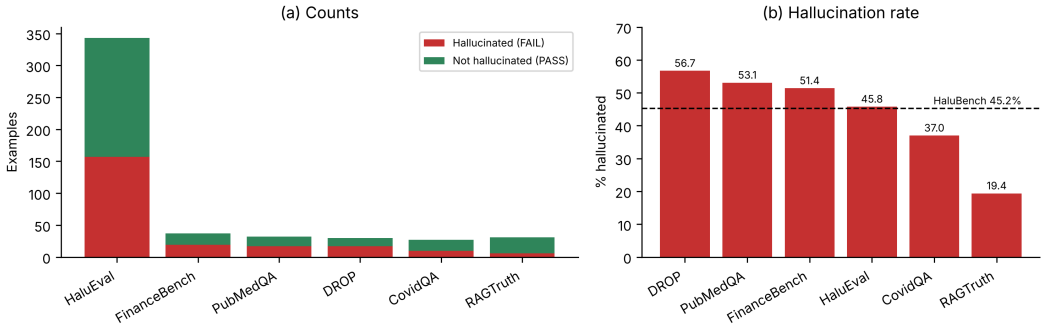


Figure 5: Hallucination labels in the 500 HaluBench rows. (a) Counts of hallucinated (FAIL) and faithful (PASS) answers per source dataset. (b) Percentage hallucinated per source, sorted, with the overall HaluBench rate of 45.2% as a dashed line; rates range from 19.4% (RAGTruth) to 56.7% (DROP).

LLM-judge pilot. We ran a small pilot to probe how a lightweight open judge behaves (Table 6 and Figure 6). On the 10 SQuAD 2.0 rows the judge correctly answered No in all 10 cases. On the 20 HaluBench rows it was cautious, answering *Unsure* on 15 and flagging 1 of 8 hallucinated

answers. The pilot also revealed a concrete, fixable cause: the HaluBench `context` field is empty in the merged file because the loader reads a field HaluBench does not use. Restoring the passages is therefore the first step towards a fair test of context-grounded judging, and the same missing-field problem appeared when the evaluation notebooks tried to re-attach them (Section 3.3). Because the pilot is small, we treat it as diagnostic and do not report an accuracy figure.

Table 6: Pilot: dataset label vs. LLM verdict on 20 HaluBench rows.

Dataset label	LLM: Yes	LLM: No	LLM: Unsure
FAIL (hallucinated)	1	2	5
PASS (faithful)	0	2	10

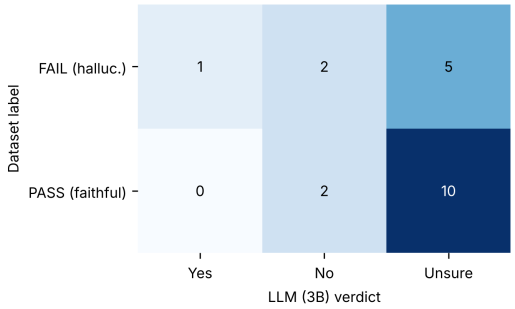


Figure 6: Pilot of the 3B LLM judge on 20 HaluBench rows: dataset label against LLM hallucination verdict. The judge answered *Unsure* on 15 of 20 rows and flagged 1 of 8 hallucinated answers.

Open LLMs as answerers. Table 7 and Figure 7 show how the four models behave on the 160 test questions. With a passage at hand, all models are strong on answerable SQuAD 2.0 questions (67.5–87.5% accuracy; Llama-3.1 is best at 87.5%), and the wrong-answer rate stays at 10–12.5%. Abstention on unanswerable questions is where the models differ: Qwen2.5 hallucinates least (15.0%), followed by Mistral (22.5%), Llama (27.5%) and Gemma (35.0%). On open-domain AmbigQA, Qwen2.5 also has the lowest wrong-answer rate on single-answer questions (15.0%), and its low accuracy (17.5%) together with that rate implies that it abstains with `UNKNOWN` on roughly two thirds of these questions, i.e., it trades coverage for caution. Gemma and Mistral are the most willing to answer: they reach the highest accuracy on single-answer questions (37.5% each) and the highest coverage of ambiguous questions (at least one valid answer for 40.0% and 52.5% of multi-answer questions), but they also have the highest wrong-answer rates (50.0% and 62.5%). Gemma flags ambiguity most often on multi-answer questions (55.0%), though it also flags 40.0% of single-answer ones; Llama is the most selective flagger (12.5% against a 2.5% false-flag rate). Mean interpretation coverage stays between 0.06 and 0.21 for all models, so covering *several* readings of an ambiguous question is hard for every model tested.

Open LLMs as judges. Table 8 and Figure 8 report the judge test. All 12 model–task runs produced parseable one-word verdicts (0% unparseable). On hallucination, Mistral (F1 0.551) and Gemma (F1 0.513) detect most of the hallucinated answers (recall 0.868 and 0.758), whereas Qwen2.5 and Llama-3.1 are conservative (recall 0.110 and 0.198). Because the HaluBench passages were not attached, these scores describe closed-book judging. The per-source breakdown (Figure 9) shows complementary strengths: Gemma and Mistral accept 80% of the gold SQuAD 2.0 answers when the passage is available (Qwen2.5 15%, Llama-3.1 20%), whereas Llama-3.1 (65%) and Qwen2.5 (60%) accept most gold AmbigQA answers (Gemma 28%, Mistral 15%), and Qwen2.5 is the best on RAGTruth (81%). For ambiguity, Qwen2.5 has the highest accuracy (0.574) and the

Table 7: Test A: models answering 40 questions per group (rates in $[0, 1]$; best value per row in bold, where a best value exists).

Metric	Llama-3.1	Qwen2.5	Gemma-2	Mistral
SQuAD answerable: accuracy ↑	**0.875**	0.800	0.800	0.675
SQuAD answerable: wrong-answer rate ↓	0.125	**0.100**	**0.100**	**0.100**
SQuAD unanswerable: hallucination rate ↓	0.275	**0.150**	0.350	0.225
AmbigQA single: accuracy ↑	0.300	0.175	**0.375**	**0.375**
AmbigQA single: wrong-answer rate ↓	0.250	**0.150**	0.500	0.625
AmbigQA single: false ambiguity flag ↓	**0.025**	0.075	0.400	0.300
AmbigQA multi: flagged ambiguous ↑	0.125	0.150	**0.550**	0.375
AmbigQA multi: mean interpretation coverage ↑	0.096	0.058	0.182	**0.213**
AmbigQA multi: ≥1 valid answer ↑	0.275	0.125	0.400	**0.525**

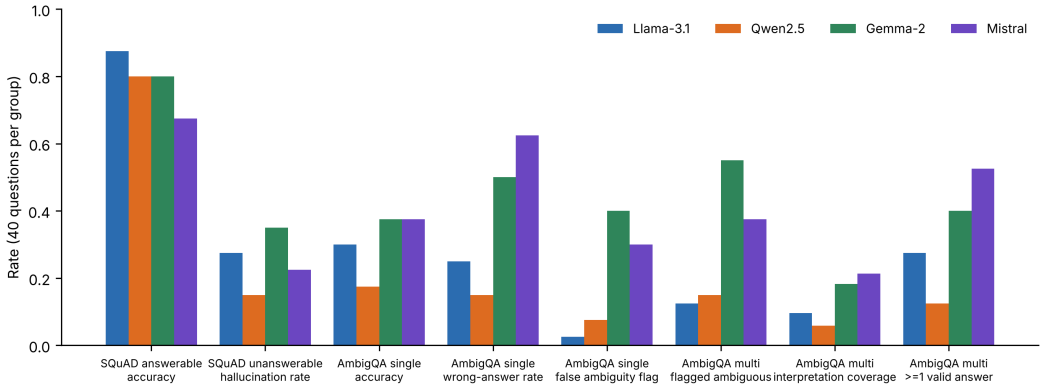


Figure 7: Test A: four open models answering 40 questions per group. Accuracy and abstention behaviour differ across models: Qwen2.5 has the lowest hallucination rate on unanswerable questions, Gemma-2 and Mistral cover the most interpretations of ambiguous questions, and Llama-3.1 is the most accurate on answerable SQuAD 2.0 questions. Lower is better for hallucination, wrong-answer and false-flag rates; higher is better for the other metrics.

most balanced precision and recall, while Gemma and Mistral reach recall 1.000 by answering Yes almost uniformly. The ambiguity test also gives independent evidence about the 3B pilot judge: on the 10 SQuAD 2.0 questions that the 3B judge had flagged as ambiguous, Gemma and Mistral disagree with it on all 10 (0.00 agreement), and Qwen2.5 and Llama-3.1 on 7 and 8 of 10 (0.30 and 0.20), which is consistent with the over-flagging we diagnosed above. For understandability, accuracy lies between 0.829 and 0.945 and F_1 for the minority "problem" class between 0.000 and 0.250. Mistral's 0.945 accuracy comes from never predicting a problem ($F_1 = 0$), so it should be compared with an always-Yes baseline rather than with chance.

5 Discussion

The corpus makes it possible to study three failure modes of QA systems (grammatical acceptability, ambiguity and hallucination) jointly on the same examples. Its labels come from complementary sources of different strength. Hallucination labels for HaluBench and ambiguity labels for AmbigQA are inherited from published annotations and form a reliable core. Treating gold SQuAD 2.0 and AmbigQA answers as not hallucinated is a transparent modelling assumption, and it supplies the negative class (226 hallucinated vs. 997 not). Since hallucinated examples come only from HaluBench, cross-source comparisons of hallucination rates should be read with source in mind; within-HaluBench comparisons are clean.

Table 8: Test B: models as judges versus corpus labels. Precision (P), recall (R) and F_1 refer to the positive class (hallucinated, ambiguous, or has a grammar/clarity problem).

Model	Task	n	Acc.	P	R	F_1
Llama-3.1	ambiguity	68	0.500	0.508	0.886	0.646
	hallucination	277	0.430	0.175	0.198	0.186
	understandability	217	0.889	0.190	0.364	0.250
Qwen2.5	ambiguity	68	0.574	0.575	0.657	0.613
	hallucination	277	0.480	0.137	0.110	0.122
	understandability	217	0.829	0.139	0.455	0.213
Gemma-2	ambiguity	68	0.529	0.522	1.000	0.686
	hallucination	277	0.527	0.388	0.758	0.513
	understandability	217	0.885	0.182	0.364	0.242
Mistral	ambiguity	68	0.529	0.522	1.000	0.686
	hallucination	277	0.534	0.403	0.868	0.551
	understandability	217	0.945	0.000	0.000	0.000

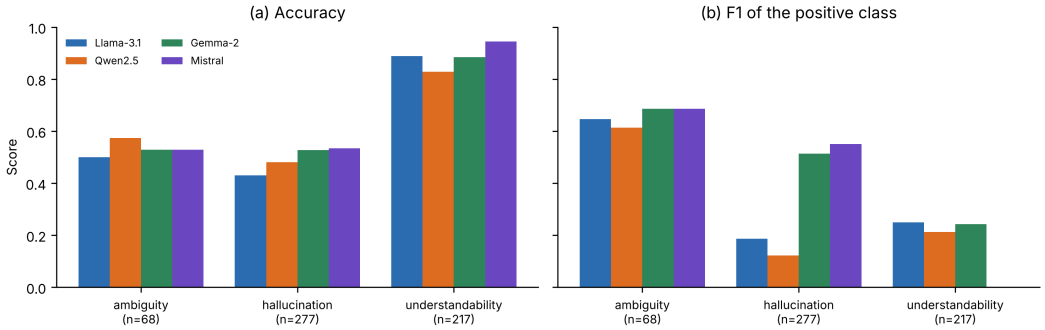


Figure 8: Test B: open models as judges of ambiguity (n=68), hallucination (n=277) and understandability (n=217). (a) Accuracy. (b) F_1 of the positive class. Hallucination F_1 is highest for Mistral (0.551) and Gemma-2 (0.513); Mistral's high understandability accuracy comes from never predicting a problem ($F_1 = 0$).

The grammar label is cheap, consistent and fully automatic. Reporting N/A for short spans protects against false negatives and highlights an interesting property of QA data: 863 of 1,500 answers are not full sentences. Our ambiguity label is deliberately broader than AmbigQA's, covering vague or competing answers as well as ambiguous questions, which opens the way to studying answer-level ambiguity alongside question-level ambiguity.

The pilot gives practical guidance for scaling up. A 3B open judge is conservative, abstaining on hallucination when context is missing and over-flagging ambiguity, which makes it a good first-pass triage tool. Two remedies are already identified: repairing the context field and moving to a larger judge (Section 3.3 evaluates four 7–9B models loaded in 4-bit), with the review flag routing uncertain rows to human annotators. The benchmark shows that the corpus is informative about the models as well as about the labels. The four models occupy different points on the cautious–willing spectrum: Qwen2.5 abstains most and hallucinates least on unanswerable questions, Gemma and Mistral answer most and cover more interpretations of ambiguous questions, and Llama-3.1 is the most accurate on answerable SQuAD 2.0 questions and the most selective flagger of ambiguity. As judges, their strengths are complementary across sources (for example, Gemma and Mistral on passage-backed SQuAD 2.0 answers, Llama-3.1 and Qwen2.5 on AmbigQA, Qwen2.5 on RAGTruth), which suggests that an ensemble with disagreement routed to humans is a promising next step. Moving from a 3B to a 7–9B judge removes the abstention problem for Gemma and

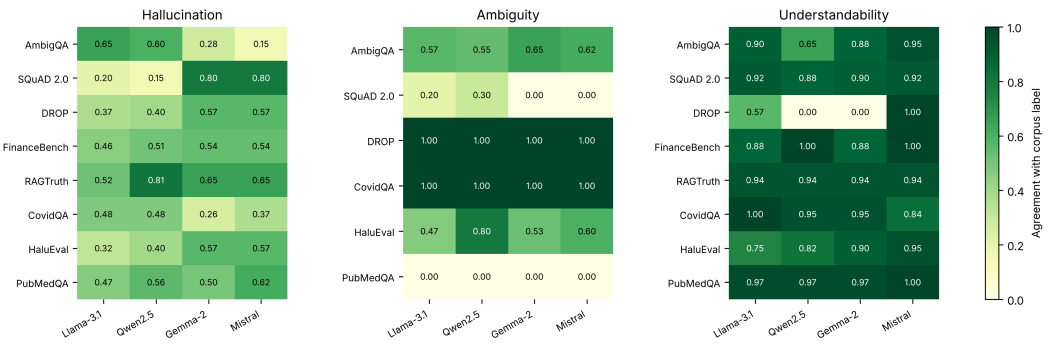


Figure 9: Per-source agreement between each judge's verdict and the corpus label (accuracy). Hallucination: Gemma-2 and Mistral accept 80% of gold SQuAD 2.0 answers when the passage is available, while Llama-3.1 and Qwen2.5 accept more gold AmbigQA answers, and Qwen2.5 is the best on RAGTruth. HaluBench rows were judged closed-book. Ambiguity cells for non-AmbigQA sources agree with labels from the 3B pilot judge and rest on very few rows (e.g. 10 SQuAD 2.0 questions), so they are indicative only.

Mistral, but closed-book hallucination judging remains difficult (best F_1 of 0.551, with accuracy of at most 0.534), so a larger judge alone is unlikely to be enough without the passages. The sample sizes also set the resolution: with 40 questions per group, one question moves a rate by 2.5 points, so differences of a few points between models should not be over-read. We view the benchmark as a first, reproducible reference point rather than a ranking. The notebook also supports a full run (`N_PER_GROUP = None`).

Scope and next steps: (i) the LLM judge has so far been run on 30 rows and will be extended to the full corpus; (ii) the manual columns are ready for human labels, which will enable inter-annotator agreement and judge-accuracy estimates; (iii) the 1,500-row English sample is a compact, reproducible starting point that can be scaled and extended to other languages; (iv) HaluBench is dominated by HaluEval-derived rows (68.6%), so rebalancing across its source datasets is a natural extension; and (v) the benchmark uses one prompt per task, one seed, 40 rows per group and 4-bit models, judges HaluBench rows closed-book, and scores ambiguity partly against labels produced by the 3B pilot judge, so the Test B numbers measure agreement with the corpus labels rather than agreement with human truth.

6 Conclusion

We presented a reproducible pipeline that merges AmbigQA, SQuAD 2.0 and HaluBench into a 1,500-example corpus labelled for grammatical correctness, ambiguity and hallucination, combining published dataset annotations with free open models. The resulting statistics (92.5% acceptable among 637 sentence-level answers, 55.4% ambiguous AmbigQA questions, 45.2% hallucinated HaluBench answers) show that the corpus spans a wide, informative range of each phenomenon. Early diagnostics have already identified a concrete fix (the HaluBench context field), and the four-model benchmark (Llama-3.1-8B, Qwen2.5-7B, Gemma-2-9B, Mistral-7B-v0.3) shows that the corpus distinguishes cautious from willing answerers and surfaces complementary judging strengths. The pipeline is set up for the next stage: running a larger judge on all rows and collecting human annotations to validate the labels and analyse how grammar, ambiguity and hallucination relate.

References

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. In *Advances in Neural Information Processing Systems*, 2023.

Dheeru Dua, Yizhong Wang, Pradeep Dasigi, Gabriel Stanovsky, Sameer Singh, and Matt Gardner. DROP: A reading comprehension benchmark requiring discrete reasoning over paragraphs. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 2368–2378. Association for Computational Linguistics, 2019.

Abhimanyu Dubey et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Gemma Team. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55, 2025.

Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. FinanceBench: A new benchmark for financial question answering, 2023.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.

Albert Q. Jiang et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577. Association for Computational Linguistics, 2019.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:452–466, 2019.

Quentin Lhoest et al. Datasets: A community library for natural language processing. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 175–184. Association for Computational Linguistics, 2021.

Junyi Li, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6449–6464. Association for Computational Linguistics, 2023.

Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252. Association for Computational Linguistics, 2022.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach, 2019.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9004–9017. Association for Computational Linguistics, 2023.

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1906–1919. Association for Computational Linguistics, 2020.

Sewon Min, Julian Michael, Hannaneh Hajishirzi, and Luke Zettlemoyer. AmbigQA: Answering ambiguous open-domain questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5783–5797. Association for Computational Linguistics, 2020.

Timo Möller, Anthony Reina, Raghavan Jayakumar, and Malte Pietsch. COVID-QA: A question answering dataset for COVID-19. In *Proceedings of the 1st Workshop on NLP for COVID-19 at ACL 2020*. Association for Computational Linguistics, 2020.

John Morris, Eli Lifland, Jin Yong Yoo, Jake Grigsby, Di Jin, and Yanjun Qi. TextAttack: A framework for adversarial attacks, data augmentation, and adversarial training in NLP. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 119–126. Association for Computational Linguistics, 2020.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, Kashun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878. Association for Computational Linguistics, 2024.

Qwen Team. Qwen2.5 technical report, 2024.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2383–2392. Association for Computational Linguistics, 2016.

Pranav Rajpurkar, Robin Jia, and Percy Liang. Know what you don't know: Unanswerable questions for SQuAD. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 784–789. Association for Computational Linguistics, 2018.

Selvan Sunitha Ravi, Bartosz Mielczarek, Anand Kannappan, Douwe Kiela, and Rebecca Qian. Lynx: An open source hallucination evaluation model, 2024.

Alex Warstadt, Amanpreet Singh, and Samuel R. Bowman. Neural network acceptability judgments. *Transactions of the Association for Computational Linguistics*, 7:625–641, 2019.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45. Association for Computational Linguistics, 2020.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*, 2023.